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Feedback-Regulated Information Fusion Approach for Optimizing Multiple Spread Control in complex Network

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Abstract

The information spread uses a social media platform, where one user intends to share information with others intentionally or non-intentionally. The rapid spread of information on social media platforms is a complex phenomenon influenced by many factors. However, the factors influencing this rapid spread in the online space still need to be discovered. Individuals utilize social media platforms to share their experiences regarding various events related to products or services. Given the intricate and multilayered nature of social media networks, the feedback received from each layer essentially mirrors the collective perceptions of the world. This process involves fusing information from each node within the network to arrive at a decision specific to the corresponding layer. And regulate the multiple spreads through the social networks. The system is verified by using a diffusion model to ensure correctness. The verification and validation show the suitability of the proposed system in controlling multiple spreading.

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Keywords: Complex networks; Social media; Information fusion; Feedback; Epidemic model

1. Introduction

A complex network has non-trivial topological features and consists of many sub systems. These interconnected subsystems are required to model a real world network [1]. The logically similar subsystems are called layers [2]. All-natural or biological or engineered complex systems that exist in the real world have multiple systems and subsystems. Interactions of metabolic systems, transportation networks, interpersonal interaction, and relationships in social networks are a few examples of the complex networks across the domain [3], [4].

Information propagation explores disseminating information, ideas, or innovations within a network and its impact on an individual's knowledge and behaviors [7]. The investigation encompasses diverse pathways through which

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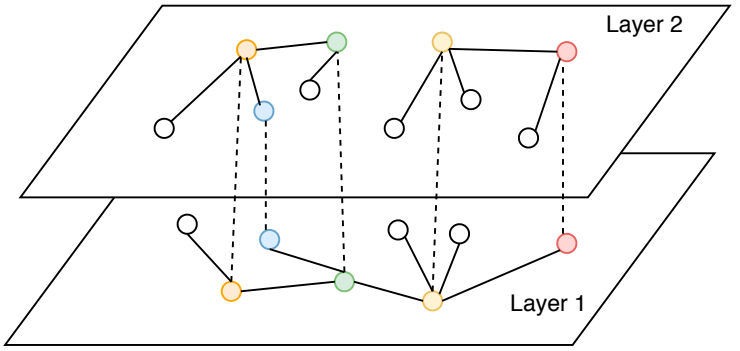


Fig. 1. A multilayer network structure having two layers, where layer 1 is the physical network modeled as small world network and layer 2 is a social network modeled as scale free network

information can circulate within intricate networks, including social media and inter-person communications. Much research has been conducted to unravel the key factors governing the information's spread and dissemination [8], [5].

In the modern age of data processing, a massive amount of data is constantly generated from diverse sources, posing a significant challenge in the scientific domain, where complex techniques are needed to extract valuable insights. This multifaceted endeavour involves various stages: data pre-processing, storage, analysis, information fusion, and retrieval. Information fusion, a crucial aspect of this process, is pivotal in refining data to generate coherent and accurate information [9]. It improves its raw state through association, amalgamation, and correlation, regardless of whether the data originates from a single source or multiple sources. Machine learning is anticipated to significantly enhance the efficiency of fusion, a method used for computation and classification [10], [11]. Online feedback is versatile in various domains, including decision-making, sentiment analysis, and epidemic control. Information fusion at the feedback and feature levels greatly enhances decision-making across diverse fields [12]. The proposed methodology combine the feedbacks and perform information fusion, which in turn controls further spreads in real world networked systems.

Major contributions of this article are:

- An information fusion extracts features from feedback and produces a decision for controlling further progress of the process and spreads of information.
- To validate the proposed methods by diffusion model.
- Analyzing the different factors affecting information spreads.

The following is a summary of the article's structure: Section 2 introduces the elementary backgrounds and summary of the literature. Section 3 describe the proposed architectural frame work for information fusion and the problem definition. The conduct of experiment, verification using an epidemic model present in Section 4, followed by results and its discussion in Section 5. Finally, the article concludes in Section 6.

2. Literature Review

Complex networks, also known as complex systems, are intricate structures comprised of interconnected elements showcasing distinctive connection arrangements and emerging properties [13]. They exist in diverse domains like social networks, the Internet, biological systems, and transportation networks. Analyzing complex networks has evolved into an interdisciplinary endeavour to uncover the fundamental rules governing their structure, behaviour, and ability to withstand challenges [14], [15]. These elaborate networks offer valuable insights into the intricate web of connections shaping the world, elucidating phenomena such as the spread of information, the transmission of diseases, the dynamics of innovation, and societal influence [6], [16].

In real-time systems, particularly those with numerous components and inter-dependencies, grasping the complex relationships among these elements can exhibit a considerable challenge. Network science provides fundamental ideas

such as small-world networks and scale-free networks [17], [18]. Unlike small-world networks, scale-free networks display a power-law distribution of node degrees, where a minority of nodes possess a notably high number of connections. In contrast, the majority of nodes have only a few. These central hubs within the network play a critical role in disseminating information and upholding connectivity. However, this resilience to random node removal renders scale-free networks susceptible to deliberate targeted attacks [19].

Information diffusion pertains to the sharing of information within a connected system by individuals or groups. It encompasses various stages, including information dissemination, propagation, and spread [20]. In practical scenarios, information permeates through different layers, and when nodes are common to these layers, their respective spreading impacts each other [21]. Nodes in each layer produce multiple instances of information spreading and associated feedback, which are combined to shape decisions that, in turn, affect the ongoing dynamics of the system [22], [23].

Information fusion is the process of enhancing the reliability and coherence of data compared to its original form [24]. It involves integrating, merging, and correlating information from one or more sources. Standard techniques for information fusion include probabilistic integration, belief-based reasoning fusion, and set-based fusion [25]. The initial fusion of raw data is called raw-level fusion, while more advanced versions are known as characteristic-level fusion. Decision-level fusion occurs when data is used to make conclusions, incorporating real-time feedback for decision-making and sentiment analysis, particularly in contexts such as epidemic control. In various fields, information fusion at the feedback and feature levels plays a crucial role in aiding decision-making [26], [27]. The representation structure of real-world systems has evolved from single-layer to multi-layer, and information fusion at each layer is vital for maintaining data consistency.

The analysis of existing literature reveals that the amalgamation of information at different stages leads to decision-making influenced by feedback from users on social media, subsequently influencing how information spreads. The suggested propagation model aids in investigating how multiple dissemination processes unfold within social networks, ultimately serving as a means to assess the effectiveness of these methods.

2.1. Background Study

The structure of the complex system is to be studied for every dynamics activities in connection with multilayer networks. All the real-world systems consist of multiple subsystems and inter-layer connectivity. Though the adjacency matrices meet the requirement for the representation of a simple networked system. But it is not adequate for representing real-world systems. So the general framework is good and that meets the challenges of multiplayer real-world systems. The elementary diffusion processes present in this section will helps the readers to understand the proposed concept in the article.

2.1.1. Mathematical Representation of Multilayer Networks

In network science, a graph is a mathematical representation of a complex network, $G = (V, E)$, where the vertex set is denoted by v, where $v = \{v_1, v_2, v_3, ..., v_n\}$. The edges are denoted by E, where $E = (v_i \times v_j); \{v_1, v_2, v_3, ..., v_n\} \in V$. A monolayer graph representation can be scaled to a multilayer graph easily [28]. A graph with multiple layers expressed as ,

$$G_m = (V_m, E_m, V, L) \tag{1}$$

where V denotes set of vertices, $|V| = N$, and L denotes the elementary layers, $|L| = m$. The layer notation, L_d^a in which d denotes number of layers and a is *aspect*. Let multilayer networks with $d = 2$ and $a = 2$ corresponds to a networks have 2 layers in each aspects. A layers in the multilayer networks, L is defined as, $L = \{L_d^a\}_{d=1}$. The notation, V_m, represents nodes in each layer, $V_m \subseteq V \times L_1 \times \ldots \times L_m$. Interconnection edges between layers expressed as E_m,where $E_M \subseteq V_m \times V_m$. The number of vertices in the layer α is expressed as $V_\alpha = \{v_1^\alpha, v_2^\alpha, .., v_{N_\alpha}^\alpha\}$ and set of edges between layer α and β is expressed as $E_m = \{E_{\alpha\beta} \subseteq V_\alpha \times V_\beta; \alpha, \beta = \{1, 2, .., m\}, \alpha \neq \beta\}$.

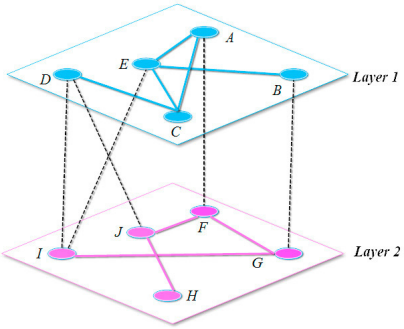


Fig. 2. An example multilayer network graph with $|V_1| = 5$, $|V_2| = 5$ and $|L| = 2$

2.1.2. Supra-Adjacency Matrix

A matrix is the elementary representation for graph in the network science. Since multilayer matrix is the stack of layers, so an aggregation of adjacency matrix of the individual layers is capable for representing such a complex graph. Therefore, individual graph G_α, is expressed as $= A^\alpha_{ij} = a^\alpha_{ij}$, where

$$a^\alpha_{ij} = \begin{cases} 1, & if \langle v_i, v_j \rangle \in E_\alpha where\ \{v_i, v_j\} \in V^\alpha \\ 0, & otherwise \end{cases} \tag{2}$$

The interconnection between layers, the edges between α and β is expressed as $E_{\alpha\beta}$, The corresponding matrix representation, $A^{[\alpha\beta]}_{dj} = (a^{\alpha\beta}_{ij})$ is given by

$$a^{\alpha\beta}_{ij} = \begin{cases} 1, & \text{if } \langle v^\alpha_i, v^\beta_j \rangle \in E_{\alpha\beta} \\ 0, & otherwise \end{cases} \tag{3}$$

If an un-directed edges exist in the networks, the adjacency matrix is a symmetric matrix or order n^α, number of nodes in the layer α. For each entry, $a^\alpha_{ij} = 1$ exist only when a link connects from j to i in $G^{[\alpha]}$. The coupling matrix, $C \leftarrow G_c = \{C_{ij}\}$ an $N \times N$ matrix with $C_{ij} = 1$ when a node present in different layers, usually in multiplex networks.

A featured matrix such as Laplacian matrix derived from the adjacency matrix, which helps to find the characteristic of the given graph. The same can be scaled to multilayer networks, the corresponding matrix is called Supra-Laplacian matrix. Let D be the diagonal matrix and A be the Supra-Adjacency matrix, the Supra-Laplacian matrix can compute by $\bar{L} = \bar{D} - \bar{A}$. Some of the properties like coupling strength in multilayer networks and diffusion properties are discovered with the help of Supra-Laplacian matrix [28], [**?**].

3. Fusion Architecture

The raw data need to be undergone a fusion process. Thus, design a hierarchical system to process the data. The fusion process is complex, where feedback from all the nodes combines at each layer and purifies it. The result of one layer is fed forward to the next level. A machine learning algorithm fuses the information produced at each layer to generate a decision [29], [30]. That helps to reduce the spread volume. The design of a fusion model for the proposed spread control is illustrated in Fig. 3.

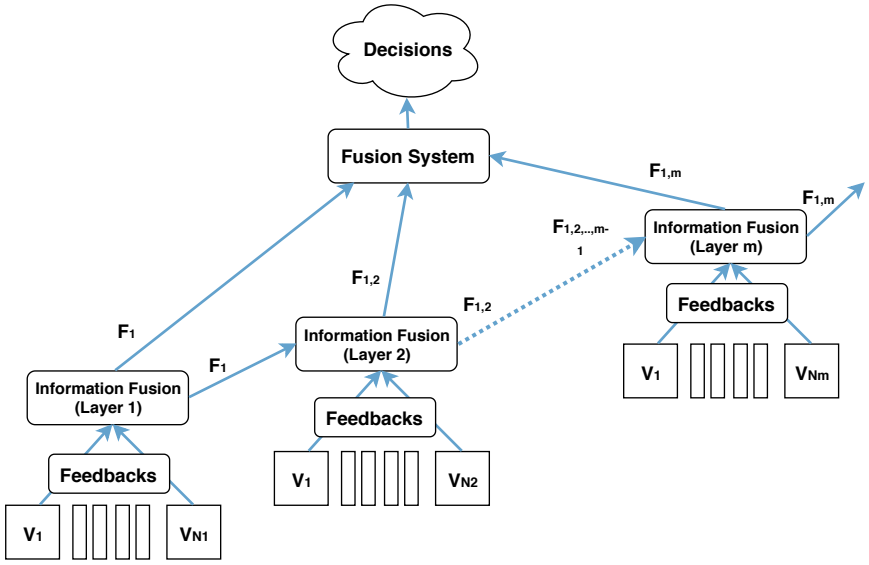


Fig. 3. Information diffusion model for controlling spread.

Proposed Information Fusion Process

Problem Description : When an information spreads over the community, people respond with their experiences. This feedback from different communities help other people to take necessary action. However, the number of online feedback increases beyond the limits by which the system can manage, it leads to an increase in the complexity of processing. In this case, a feedback analysis system helps to extract an exact information, generally a decision. The decision can be made possible by information fusion □.

The information fusion works on either complex data or soft data. Data provided by humans are generally called soft data. All the fusion methods used for fusing soft data. The operation performed during the fusion process is to fuse similar features. The weighted average is used for ranking the feedback for producing a decision. The block diagram for the information fusion process at each layer is depicted in Fig. 4.

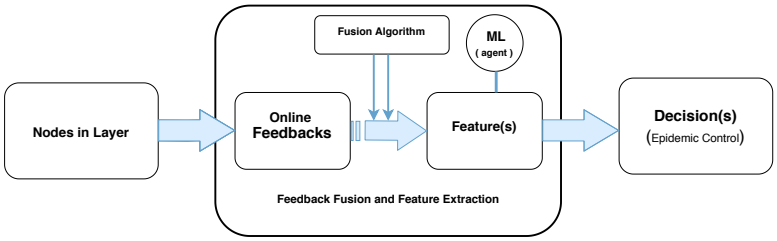


Fig. 4. Extraction and ranking of features of the dynamics of spread from feedback by information fusion process

The fusion process carries out through the filtration and purification of its evaluation. It carries out either partial supervision or without supervision. The unsupervised machine learning algorithms are best suitable for the information fusion process. Here, an unsupervised machine learning algorithm to assign a rank to the features of the epidemic. A k-means clustering algorithm uses the optimizing on the identification of the features. The algorithm process the features and grouped into categories [29]. The simulation results show that the k-means algorithm gives the solution for ranking and clustering features. Information from multiple sources or the same source at different intervals of time is purified and filtered its characteristic features. Epidemic control decisions produce either automatically or human intervention. The fusion revamps the information with more quality, consistency, accuracy, and reliability.

The online feedback are generated by web crawling and result in the form of text data and pre-processed. All the features extracted in the first phase. During the second phase, the strength of each feature is identified and ranked based on machine learning [31]. The collection of each feature with a score is produced in the second phase. The result of the third phase produced according to the previous phases, and the decision is made available to people by information fusion.

4. Analysis of Regulating Parallel Diffusion by Epidemic Model

In an epidemic model, the diffusion process occurs among nodes, which can be part of the same layer (inter-layer) or different layers (intra-layer). Information spreads from an initial subset of nodes, and these nodes consistently propagate the information to neighbouring nodes. This diffusion process primarily employs the SIS/SIR model for most of the spreading phase, where infected nodes eventually recover, resembling the fundamental epidemic model [32]. This study employs an epidemic model to investigate how information spreads and multiple parallel processes interact in complex networks. It focuses on various layers where processes occur concurrently, each affecting the others. The research utilizes the *Awareness Regulated Spreading (ARS)* model, where one process propagates within a layer, and its actions, guided by feedback, spread across another layer, a phenomenon commonly observed in social networks.

Algorithm 1 Selection of seed from information layer (social networks)

Input: $G^{\alpha} = (V^{\alpha}, E^{\alpha})$: *A social graph*; C: *Community structure*; k: *A positive number*;
Output: S:*seed*

procedure EpiFast($G, Seed, C, k$)
  $seed \leftarrow null$,
  $flag(node) \leftarrow 0$
  while *Each node*, $i \leq N$ **do**
    $\delta(u) = 1$
    $T_S(u) = \sum_{v \in N^+(u)} p_{uv}$
  for each node $i \leftarrow$ 1 *to* C **do**
    for each node $j \leftarrow N(i)$ *to* $d(i)$ **do**
      while $|Seed| < k$ **do**
        $v \leftarrow argMin_{u \in V/S, u \in V-S}(S + C_i(u)) - seed)$
        $seed \leftarrow seed \cup \{u\}$
      $\delta(w) \leftarrow (1 - p_{uv})\delta(w)$
  return $seed, S$

The crucial factor for assessing the spread of an epidemic is the number of infected nodes or the epidemic volume. The algorithm for validation process is shown in the Algorithm 1. The epidemic threshold significantly impacts this validation process, which expands its scope. The rate at which awareness spreads plays a vital role in determining the critical epidemic threshold, as observed in the ARS model, where a higher level of awareness among nodes raises the threshold and subsequently alters the spread dynamics. To comprehensively evaluate the impact of the ARS model, a quantitative analysis was conducted to measure the significance of the awareness rate on the total epidemic volume (the number of infected nodes). The analysis reveals that the spread decreases as the threshold increases, proving that the system's transmissibility is minimized.

The worst-case complexity is computed as $O(|V| + |E| + LM_c)$ where $L = clusters$ and M_c is the number of nodes in each community. And space complexity is $O(|V| + |E|)$. Let us consider a graph with $|V|$ number of nodes and $|E|$ number of edges, the algorithm requires $O(|V|)$ and $O(|E|)$ steps respectively. In the Algorithm 1, the maximum cut requires $O(|N^+(v)|)$ time. It seems to be the total number of nodes in the given cluster. In the worst case, it is $O(|M_c|)$. There are L such clusters, so time complexity is $O(kM_C)$. The algorithm recursively executes and find k nodes as seed. The worst-case time complexity is $O(|V| + |E| + LM_c)$. It approximate as $O(k * |E|)$. Since there are $|V|$ nodes and $|E|$ edges to be traversed, the space complexity is $O(|V| + |E|$.

5. Result and Discussion

The validation process is carried out to fix its effectiveness in controlling the information diffusions by simulations. Let us consider the situations where multiple spreading processes exist in different layers. The experiment was conducted with three real dataset for the experiments: the epidemic spreads physically, and information about prevention and awareness spreads in the social network layer. The objective of the simulation is to find the epidemic threshold. As the value is higher for the epidemic threshold, the spread decreases. As awareness spread increases in the social network, the epidemic spread decreases in the physical networks. The simulator is developed using Python 3.6.0 and its supporting package, NetworkX 2.4. The more layers are added to the networks, the more computational complexity increases.

Dataset

- *higgs multiplex*: It is a two-layer multiplex network with 4,56,631 nodes and each edges associated with a weight. The first layer is a social network, where the interactions in Twitter corresponding to the different actions are taking place in the social networks [33]. The Higgs dataset has created after observing the spreading process on Twitter.
- *ff-tw*: This dataset has been made known to the public in the year 2011 by Magnani and Rossi [34]. The dataset supports the users to post messages and comments for other messages similar to Facebook by the way through another system. There are 3,22,697 users so far and 15,87,273 services are available in the network. But consider only two layers from them and users who have registered on Twitter. It is a two-layer multiplex network with 1,55,804 nodes and 13657550 undirected and unweighted ties.
- *friendfeed-ita*: This dataset has public interaction between nodes of the Friendfeed social network over two months [34]. It is defined as three layers in which physical contact and social interaction layers above it. There are 21,006 nodes and 5,73,600 links between nodes exist in the *ff-tw* network. It is a subset of the networks, information about Italian users.

The experiments conducted analytically and investigate the effects of multiple spreading in real complex networks. The analysis is performed by computing the below mentioned parameters: degree of immunization and epidemic threshold.

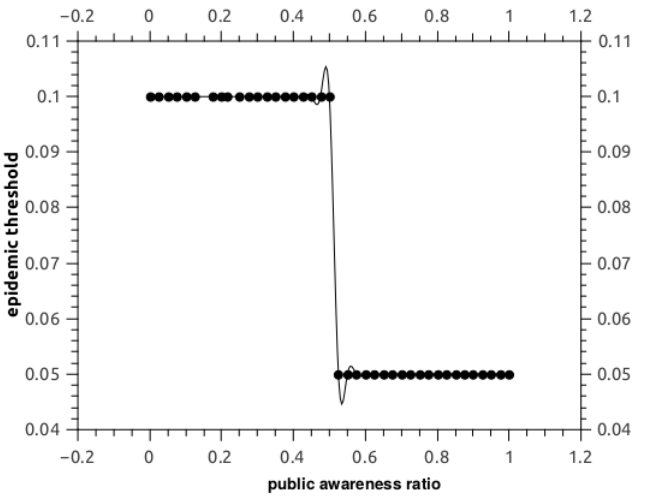


Fig. 5. The figure shows the relationship between the epidemic threshold and awareness rate, There is a sharp change occur when awareness rate reaches 0.5. The epidemic threshold drastically reduces when awareness rate reaches 0.5. The transition from $0 \leq 0.5$, the threshold remain constant, there is an abrupt transition when $\alpha \geq 0.5$.

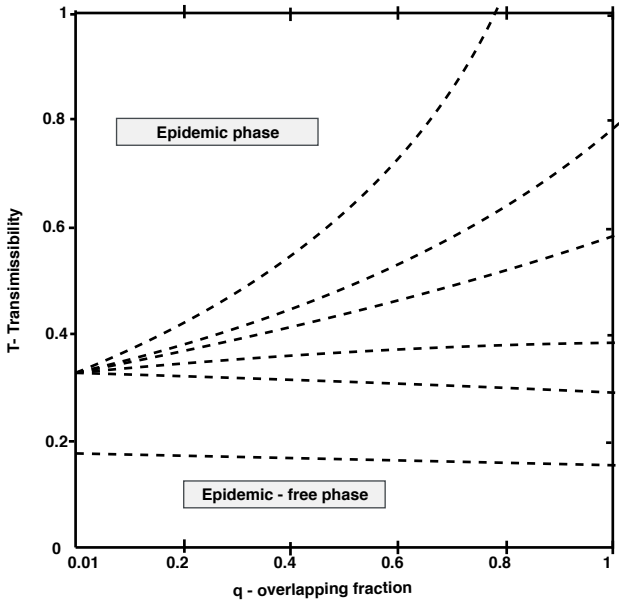


Fig. 6. The statistical correlation of dependence epidemic threshold and overlapping fraction for the eidemic model in multilayer networks. The phase diagram of T-q for a multilayer network with aspect=2 and line graph corresponds to transmissibility obtained theoretically. The overlapping fraction, $q \leftarrow 0$ corresponds to layers are isolated and $q \leftarrow 1$ represents the fully connected multiplex networks. The epidemic-free phase and the region above T correspond to the epidemic-phase are well defined.

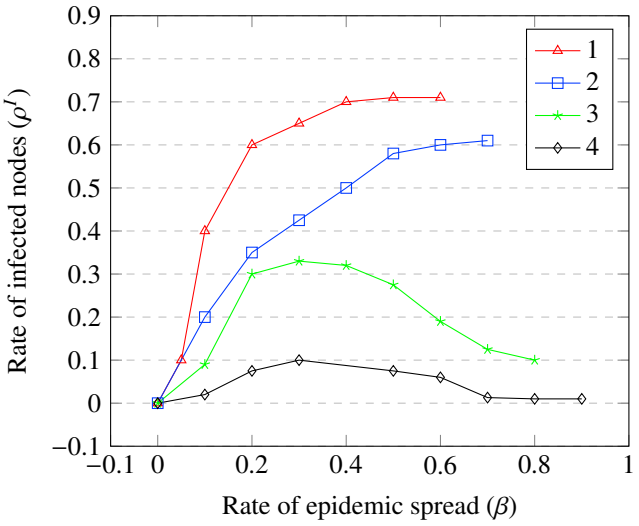


Fig. 7. With different values of awareness rate, the curves display the effect of the final epidemic volume. The different awareness rate for the label, (1, 0), (2, 0.25), (3, 0.5) and (4, 0.75).

The experiment was repeated with different awareness rate values and labelled 1, 2, 3, and 4 as 0.00, 0.25, 0.5, and 0.75 respectively. The higher infection ratio and significant awareness rate result in more individuals becoming aware, and each individual turns to boost the spreading of awareness. The coupled dynamic process in multilayer networks benefited from knowing the final epidemic size with different awareness rates, as shown in Fig. 7. The volume of individual entities infected in a network determines the final epidemic volume, as the epidemic volume gets smaller when the value of the awareness rate increases.

6. Conclusion

In this article, The experiments carried out an information fusion process where various feedback at each layer is combined to produce features of the epidemic spread. The k-means algorithm is applied to train the features and produce the decision. The proposed framework for information fusion found efficient consistent throughout the process. The system was trained in a way that helped to scale down the epidemic volume. The study validates the fusion procedure using an epidemic model and investigates the dynamics of epidemic spread in multilayer networks with online feedback. Various values of epidemic threshold and awareness rate are explored successfully. The transmission of control information occurs within social networks, while immunization and eradication processes occur in physical networks, with communication between them happening on the same nodes across multiple layers. The immediate impact of social media on physical networks is evident due to concurrent processes. The epidemic control dissemination rate falls within the range of 0.00 to 0.50, leading to different final epidemic sizes depending on the epidemic threshold, which increases with lower awareness rates and can reach a maximum of 1, resulting in epidemic outbreaks.

Moreover, the experiments yielded valuable insights into mitigating epidemic outbreaks and offered actionable preventative measures. Hence it is observed that the efficacy of immunization rises in tandem with the likelihood of interaction intensity. Furthermore, the epidemic management algorithm demonstrates its utility across various domains, including but not limited to curbing the dissemination of rumors, controlling the spread of malware, and managing epidemics.

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References

[1] *Newman, Mark. Networks. Oxford university press, 2018.*
[2] *Newman, Mark, Albert-László Barabási, and Duncan J. Watts. The structure and dynamics of networks. Princeton university press, 2011.*
[3] *Estrada, Ernesto, Michele Benzi, Guanrong Chen, Peter Csermely, Jennifer Dunne, Neil Johnson, Tamara G. Kolda et al. "Journal of Complex Networks." (2013).*
[4] *Boccaletti, Stefano, Ginestra Bianconi, Regino Criado, Charo I. Del Genio, Jesús Gómez-Gardenes, Miguel Romance, Irene Sendina-Nadal, Zhen Wang, and Massimiliano Zanin. "The structure and dynamics of multilayer networks." Physics reports 544, no. 1 (2014): 1-122.*
[5] *Gayathri, R. G., and Jyothisha J. Nair. "ex-FTCD: A novel mapreduce model for distributed multi source shortest path problem." Journal of Intelligent Fuzzy Systems 34, no. 3 (2018): 1643-1652.*
[6] *Albert, Réka, and Albert-László Barabási. "Statistical mechanics of complex networks." Reviews of modern physics 74, no. 1 (2002): 47.*
[7] *Zhang, Zi-Ke, Chuang Liu, Xiu-Xiu Zhan, Xin Lu, Chu-Xu Zhang, and Yi-Cheng Zhang. "Dynamics of information diffusion and its applications on complex networks." Physics Reports 651 (2016): 1-34.*
[8] *Jisha, R. C., P. S. Indrajith, and S. Abhishek. "Community detection using graph partitioning." In 2021 2nd global conference for advancement in technology (GCAT), pp. 1-6. IEEE, 2021.*
[9] *Fan, Zhi-Ping, Guang-Ming Li, and Yang Liu. "Processes and methods of information fusion for ranking products based on online reviews: An overview." Information Fusion 60 (2020): 87-97.*
[10] *Meng T, Jing X, Yan Z, Pedrycz W. A survey on machine learning for data fusion. Information Fusion. 2020 May 1;57:115-29.*
[11] *Rani, S. Siji, and Sangeeth Sajan Baby. "Real-time Influencer Detection In Twitter Using A Hybrid Approach." Procedia Computer Science 215 (2022): 461-470.*
[12] *Camacho, David, Angel Panizo-LLedot, Gema Bello-Orgaz, Antonio Gonzalez-Pardo, and Erik Cambria. "The four dimensions of social network analysis: An overview of research methods, applications, and software tools." Information Fusion 63 (2020): 88-120.*
[13] *Boccaletti, Stefano, Vito Latora, Yamir Moreno, Martin Chavez, and D-U. Hwang. "Complex networks: Structure and dynamics." Physics reports 424, no. 4-5 (2006): 175-308.*
[14] *Nair, Lekshmi S., Jo Cheriyan, and J. Swaminathan. "Microscopic structural analysis of complex networks: An empirical study using motifs." IEEE Access 10 (2022): 33220-33229.*
[15] *Nair, Lekshmi S., Swaminathan Jayaraman, and Sai Pavan Krishna Nagam. "An improved link prediction approach for directed complex networks using stochastic block modeling." Big Data and Cognitive Computing 7, no. 1 (2023): 31.*
[16] *Estrada, Ernesto. The structure of complex networks: theory and applications. Oxford University Press, USA, 2012.*
[17] *Barabási, Albert-László, and Eric Bonabeau. "Scale-free networks." Scientific american 288, no. 5 (2003): 60-69.*
[18] *Amaral, Luis A. Nunes, Antonio Scala, Marc Barthelemy, and H. Eugene Stanley. "Classes of small-world networks." Proceedings of the national academy of sciences 97, no. 21 (2000): 11149-11152.*

[19] *Barabási, Albert-László. "Scale-free networks: a decade and beyond." science 325, no. 5939 (2009): 412-413.*
[20] *Li, Mei, Xiang Wang, Kai Gao, and Shanshan Zhang. "A survey on information diffusion in online social networks: Models and methods." Information 8, no. 4 (2017): 118.*
[21] *Santhosh, Nikita Mariam, Jo Cheriyan, and Lekshmi S. Nair. "A multi-model intelligent approach for rumor detection in social networks." In 2022 International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS), pp. 1-5. IEEE, 2022.*
[22] *Jalali, Mohammad S., Armin Ashouri, Oscar Herrera-Restrepo, and Hui Zhang. "Information diffusion through social networks: The case of an online petition." Expert Systems with Applications 44 (2016): 187-197.*
[23] *Cheriyan, Jo, and Jyothisha J. Nair. "Influence Minimization With Node Surveillance in Online Social Networks." IEEE Access 10 (2022): 103610-103618.*
[24] *Valet, Lionel, Gilles Mauris, and Philippe Bolon. "A statistical overview of recent literature in information fusion." In Proceedings of the Third International Conference on Information Fusion, vol. 1, pp. MOC3-22. IEEE, 2000.*
[25] *Salcedo-Sanz, Sancho, Pedram Ghamisi, María Piles, Martin Werner, Lucas Cuadra, A. Moreno-Martínez, Emma Izquierdo-Verdiguier, Jordi Muñoz-Marí, Amirhosein Mosavi, and Gustau Camps-Valls. "Machine learning information fusion in Earth observation: A comprehensive review of methods, applications and data sources." Information Fusion 63 (2020): 256-272.*
[26] *Blasch, Erik, Élio Bossé, and Dale A. Lambert. High-level information fusion management and systems design. Artech House, 2012.*
[27] *Lambert, Dale A. "Grand challenges of information fusion." In Proceedings of the 6th International Conference on Information Fusion, vol. 1, pp. 213-220. 2003.*
[28] *Estrada, Ernesto, Michele Benzi, Guanrong Chen, Peter Csermely, Jennifer Dunne, Neil Johnson, Tamara G. Kolda et al. "Journal of Complex Networks." (2013).*
[29] *Coates, Adam, and Andrew Y. Ng. "Learning feature representations with k-means." In Neural Networks: Tricks of the Trade: Second Edition, pp. 561-580. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012.*
[30] *Modha, Dharmendra S., and W. Scott Spangler. "Feature weighting in k-means clustering." Machine learning 52 (2003): 217-237.*
[31] *Dessi, Andrea, and Maurizio Atzori. "A machine-learning approach to ranking RDF properties." Future Generation Computer Systems 54 (2016): 366-377.*
[32] *Nair, Lekshmi S., and Jo Cheriyan. "SIR-M Epidemic Model: A SARS-CoV-2 Perspective." In Computational Intelligence and Data Analytics: Proceedings of ICCIDA 2022, pp. 499-508. Singapore: Springer Nature Singapore, 2022.*
[33] *De Domenico, Manlio, Antonio Lima, Paul Mougel, and Mirco Musolesi. "The anatomy of a scientific rumor." Scientific reports 3, no. 1 (2013): 2980.*
[34] *Dickison, Mark E., Matteo Magnani, and Luca Rossi. Multilayer social networks. Cambridge University Press, 2016.*